

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I M.Kyathi Sai Priya (192572101) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Kyathi

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Ans : Smart Retail Inventory Management System

Introduction

A Smart Retail Inventory Management System is a cloud-based solution designed to manage inventory, sales, and product information across multiple retail stores from a centralized platform. Traditional inventory systems store data locally, making it difficult to synchronize information between branches. In contrast, a cloud-based system stores all data in cloud servers, allowing every store to access updated information in real time. Devices such as Point of Sale (POS) systems, barcode scanners, mobile inventory devices, and management dashboards communicate with cloud services through Application Programming Interfaces (APIs). This enables retailers to monitor stock levels, process sales, generate reports, and predict future product demand efficiently while reducing operational costs and improving customer satisfaction.

Client Infrastructure and API Communication

The client infrastructure consists of all the devices and applications used by employees to interact with the inventory system. These include POS terminals used for billing, barcode scanners for product identification, mobile inventory applications used by warehouse staff, and web-based dashboards used by managers. Whenever a customer purchases a product, the POS system records the transaction and sends the information to centralized cloud services using secure APIs. Similarly, when a store manager checks inventory or generates sales reports, the dashboard sends API requests to retrieve the required information from the cloud. APIs act as communication bridges between client devices and cloud services, ensuring that all stores receive real-time updates. This approach improves accuracy, eliminates duplicate data entry, and allows different software applications to communicate seamlessly regardless of the operating system or programming language.

Cloud-Based Inventory Management

The core of the system is the cloud-based inventory management service, which maintains accurate records of all products across multiple retail locations. Whenever a sale is completed, the inventory count is automatically reduced in the cloud database. Similarly, when new products are received from suppliers, the stock quantity is immediately updated. Since all stores share the same centralized database, inventory information remains synchronized across every branch. Employees can instantly check product availability, transfer stock between stores, or identify items that need replenishment. This centralized management significantly reduces inventory errors, prevents stock shortages, minimizes overstocking, and improves operational efficiency.

Cloud Storage and Data Management

Cloud storage plays a vital role in the system by storing all business-related information securely. The cloud database maintains product details such as product ID, name, category, price, supplier information, and stock quantity. It also stores customer purchase records, sales transactions, inventory logs, supplier details, employee information, and business analytics datasets. Every transaction performed within the system is recorded in transaction logs, which help administrators monitor user activities, identify system errors, and perform security audits. Because data is stored in the cloud, it can be accessed from any authorized location, and automatic backups ensure that valuable business information is protected against hardware failures or disasters.

Platform Services for Analytics and Demand Forecasting

The inventory management system uses Platform as a Service (PaaS) to provide advanced analytics and demand forecasting capabilities. The analytics platform collects sales data from all retail stores and analyzes customer purchasing patterns, seasonal trends, promotional campaigns, and historical sales records. Using Artificial Intelligence (AI) and Machine Learning (ML) algorithms, the system predicts future product demand. For example, if sales of a particular product increase significantly during festival seasons, the forecasting service recommends increasing stock before the next festival. These predictions help retailers avoid stock shortages, reduce excess inventory, optimize warehouse utilization, and improve overall profitability. Managers can also view interactive dashboards displaying sales performance, inventory status, and customer trends to support better business decisions.

Cloud Service Models Used

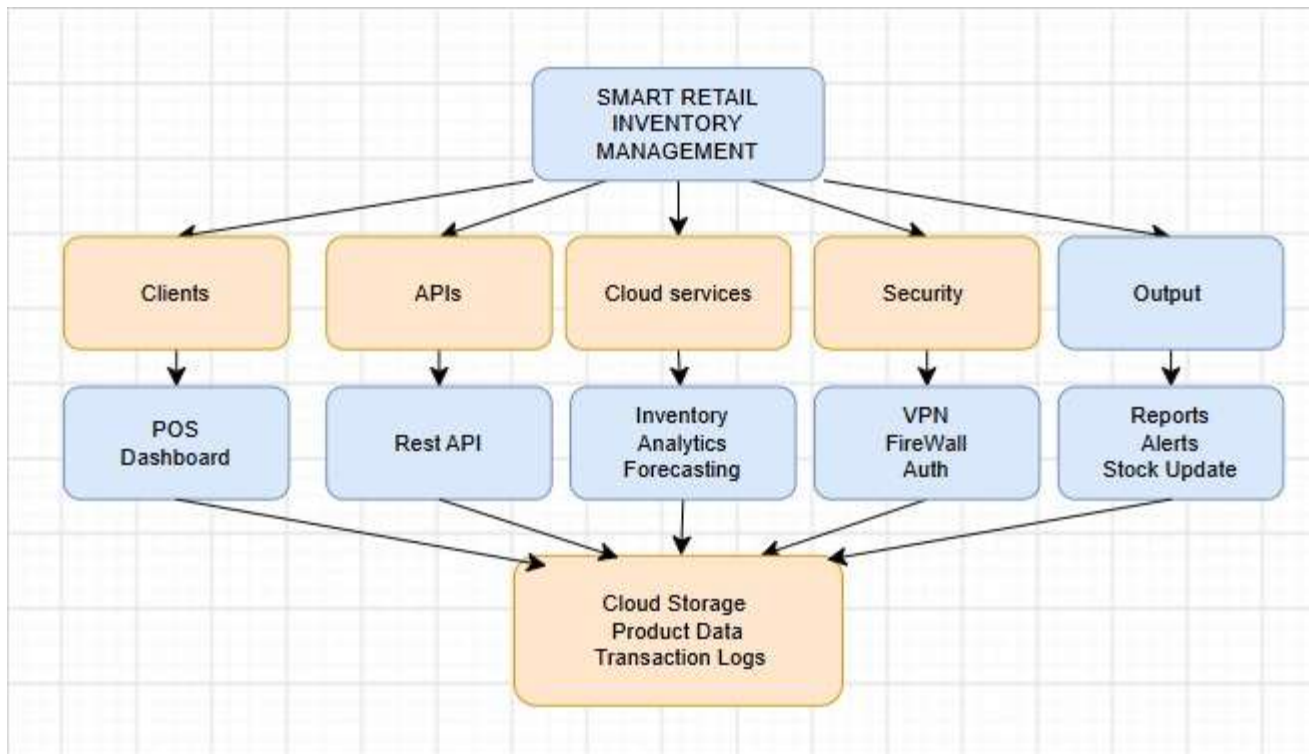
The Smart Retail Inventory Management System utilizes all three major cloud service models. **Infrastructure as a Service (IaaS)** provides virtual machines, storage, networking, and computing resources required to host the cloud applications. These resources can be scaled automatically as business demand increases. **Platform as a Service (PaaS)** offers software development tools, database management services, analytics platforms, and machine learning environments that enable developers to build and deploy advanced inventory applications without managing hardware. **Software as a Service (SaaS)** delivers the inventory management software directly to users through web browsers or mobile applications, allowing employees to access the system without installing complex software on their local devices. Together, these service models provide flexibility, scalability, and cost-effective cloud operations.

Security Mechanisms

Since the inventory management system handles sensitive business information, multiple security measures are implemented to protect data and network communication. A **Virtual Private Network (VPN)** establishes secure encrypted connections between retail stores and cloud servers, preventing unauthorized interception of data during transmission. **Firewalls** continuously monitor incoming and outgoing network traffic, blocking malicious requests and protecting the system from cyberattacks. An **API Gateway** serves as a secure entry point for all API requests by authenticating users, verifying request validity, controlling traffic flow, and preventing unauthorized access. Additionally, user authentication mechanisms such as usernames, passwords, Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC) ensure that employees can access only the resources required for their job responsibilities. These security mechanisms collectively maintain confidentiality, integrity, and availability of business data.

System Workflow

The operation of the Smart Retail Inventory Management System begins when a customer purchases a product at the checkout counter. The POS system records the transaction and sends the purchase details to the cloud server through an API. The cloud application validates the request and updates the centralized inventory database by reducing the available stock quantity. Simultaneously, the transaction details are stored in cloud storage, and the analytics service updates sales reports and performance metrics. If the inventory level falls below a predefined threshold, the system automatically generates alerts or purchase orders for suppliers. Managers can immediately view updated inventory levels and sales reports through cloud dashboards, ensuring real-time decision-making and efficient control.



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2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Ans : Cloud-Based Document Collaboration System

Introduction

A Cloud-Based Document Collaboration System is an online platform that allows multiple users to create, edit, share, and manage documents simultaneously from different locations. Applications such as Google Docs have transformed the way organizations collaborate by eliminating the need to exchange multiple copies of the same document through email. Instead, documents are stored in the cloud and accessed through a web browser or mobile application. Every modification made by one user is instantly synchronized with all other users working on the same document. This enables real-time collaboration, improves productivity, reduces duplication of work, and ensures that everyone always works on the latest version of the document.

Client Access through Web Browsers

Users access the document collaboration platform using standard web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari without installing complex software. After logging into the system using their authenticated credentials, users can create new documents, open existing files, edit content, insert images or tables, comment on documents, and share files with other users. Since the application runs through a browser, users can access their documents from desktops, laptops, tablets, or smartphones regardless of the operating system. This browser-based access improves flexibility and allows employees to work remotely from any location with an internet connection.

Real-Time Collaboration and Editing

One of the most important features of the system is real-time collaborative editing. Multiple users can work on the same document simultaneously, and every change is reflected instantly on all connected devices. For example, if one employee edits a paragraph while another corrects grammar or inserts images, both changes appear immediately without requiring users to manually refresh the document. The collaboration system manages simultaneous edits using synchronization techniques and conflict resolution algorithms to prevent data loss. Users can also communicate through comments, suggestions, and collaborative editing tools, making teamwork more efficient and reducing project completion time.

Distributed Cloud Storage and Version Control

All documents are stored securely in distributed cloud storage rather than on local computers. Distributed storage means that document copies are stored across multiple cloud servers located in different geographical regions. This approach improves reliability, fault tolerance, and availability because if one server fails, another server can continue providing access to the documents without interruption. The system also maintains version control, which automatically records every modification made to a document. Users can view the complete edit history, identify who made each change, compare different versions, and restore previous versions whenever necessary. Version control prevents accidental data loss and supports effective collaboration among multiple users.

Web APIs and Synchronization

The collaboration platform uses Web APIs (Application Programming Interfaces) to synchronize documents across different clients and devices. Whenever a user edits a document, the browser sends the updated information to cloud servers through secure APIs. The cloud application processes the changes and

immediately distributes them to all connected users. APIs also support functions such as document sharing, user authentication, notifications, comments, and file management. Because of API-based communication, users experience seamless synchronization whether they are working from a desktop computer, laptop, tablet, or smartphone. This ensures that everyone always views the latest version of the document.

Cloud Infrastructure and Network Performance

The cloud-based collaboration system is hosted on scalable cloud infrastructure that includes virtual servers, distributed storage systems, load balancers, and content delivery networks (CDNs). The network infrastructure is designed to provide low latency and high availability for users around the world. Load balancers distribute user requests across multiple servers to prevent overloading and maintain consistent performance during periods of heavy traffic. Data centers located in different geographical regions reduce the distance between users and servers, minimizing response time. Automatic scaling allows the system to allocate additional computing resources whenever the number of active users increases, ensuring uninterrupted service even during peak usage.

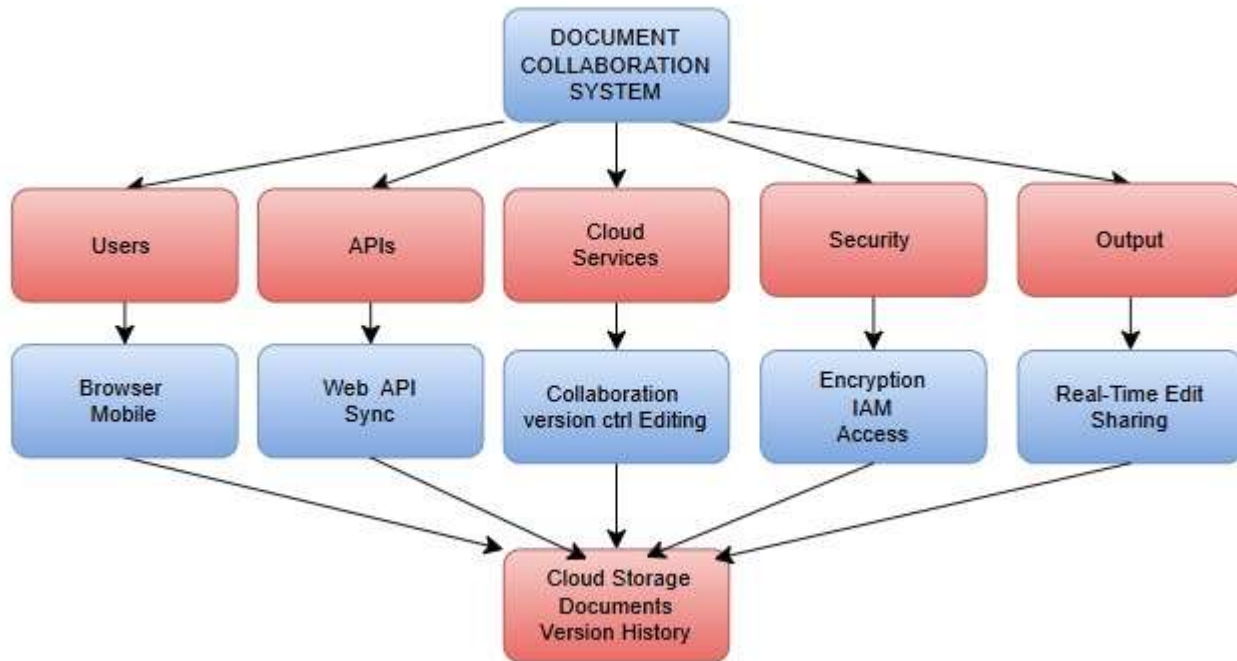
Security Mechanisms

Since business documents often contain confidential information, the collaboration system implements multiple security mechanisms to protect data. **Encryption** secures documents both while they are stored in the cloud (encryption at rest) and while they are transmitted over the internet (encryption in transit). This prevents unauthorized users from reading sensitive information. **Identity Management** ensures that only authenticated users can access the platform through secure login methods such as usernames, passwords, Multi-Factor Authentication (MFA), and Single Sign-On (SSO). The system also uses **Role-Based Access Control (RBAC)** and access permissions to control what each user can do with a document. For example, a document owner may grant another user permission to view only, comment, or edit the document. These security measures protect confidentiality, integrity, and availability while enabling safe collaboration.

System Workflow

The workflow begins when a user logs into the collaboration platform through a web browser. The user creates a new document or opens an existing one stored in the cloud. As the user edits the document, every change is immediately transmitted to cloud servers through Web APIs. The cloud application processes these updates, saves them in distributed cloud storage, records them in the version history, and synchronizes the latest changes with all other users currently editing the document. If another user joins the collaboration session, they instantly receive the latest version of the document. Throughout the process, encryption

protects data transmission, while identity management and access permissions ensure that only authorized users can access or modify the document.



3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Ans : Online Food Ordering Platform

Introduction

An Online Food Ordering Platform is a cloud-based application that allows customers to browse restaurant menus, place food orders, make online payments, and track deliveries through web browsers or mobile applications. Cloud computing enables the platform to serve thousands of users simultaneously while

ensuring fast response times, high availability, and secure transactions. The system integrates frontend applications, backend cloud services, RESTful Web APIs, cloud storage, virtual machines, and managed Kubernetes to provide a scalable and reliable food ordering experience. This architecture allows restaurants to manage orders efficiently while providing customers with real-time updates and convenient online services.

Frontend Clients and User Interaction

The frontend of the system consists of web applications and mobile applications that provide an interactive interface for customers, restaurant owners, and delivery personnel. Customers can register, log in, browse restaurant menus, search for food items, place orders, make online payments, and monitor delivery status. Restaurant managers use separate dashboards to update menus, prices, food availability, and incoming orders, while delivery personnel access mobile applications to receive delivery assignments and update delivery status. Since the applications are cloud-based, users can access the platform anytime and from any device connected to the internet, improving convenience and accessibility.

Backend Services and RESTful Web APIs

The backend of the food ordering platform contains the business logic responsible for processing customer requests and managing system operations. Frontend applications communicate with backend services using RESTful Web APIs, which enable secure and efficient data exchange between clients and cloud servers. Whenever a customer selects food items and places an order, the frontend sends an API request to the backend. The backend validates the request, checks menu availability, calculates the total bill, processes payment information, stores the order in the database, and sends confirmation back to the customer. RESTful APIs also support user registration, login, restaurant management, payment processing, order tracking, notifications, and customer feedback. This API-based communication allows different client applications to work seamlessly with the same cloud backend.

Cloud Storage for Data Management

Cloud storage plays a critical role in storing and managing various types of application data. Menu images uploaded by restaurants are stored securely in cloud object storage, allowing customers to view high-quality images without affecting application performance. Customer information, including account details, addresses, order history, and payment records, is securely maintained in cloud databases. Digital invoices and payment receipts are also stored in cloud storage, enabling customers to access them whenever required. Automatic backups provided by cloud storage protect business data from accidental loss and hardware

failures. Since cloud storage is scalable, the platform can efficiently handle increasing amounts of customer data and multimedia files as the business grows.

Cloud Infrastructure with Virtual Machines and Kubernetes

The online food ordering system is deployed on cloud infrastructure that uses **Virtual Machines (VMs)** and **Managed Kubernetes** to ensure scalability, reliability, and efficient resource utilization. Virtual machines provide computing resources such as processors, memory, storage, and networking required to run backend applications. Managed Kubernetes orchestrates multiple application containers by automatically deploying, monitoring, and scaling them according to user demand. During peak hours, such as lunch or dinner time, Kubernetes automatically creates additional application instances to handle the increased number of customer requests. When traffic decreases, unnecessary resources are released automatically, reducing operational costs. This dynamic scaling ensures continuous application availability and excellent performance even during heavy workloads.

Security Mechanisms

Security is a fundamental requirement for an online food ordering platform because it processes personal information, payment details, and business data. The system uses **authentication tokens**, such as JSON Web Tokens (JWT), to verify user identity after successful login. Each API request includes a valid authentication token, allowing only authorized users to access protected resources. All communication between clients and cloud servers occurs over **HTTPS**, which encrypts transmitted data and protects it from unauthorized interception. The platform also implements **Role-Based Access Control (RBAC)** to provide different access privileges for customers, restaurant managers, delivery personnel, and administrators. For example, customers can place orders and view their own order history, restaurant managers can update menus and manage restaurant orders, delivery personnel can update delivery status, and administrators have full control over the entire platform. These security mechanisms ensure confidentiality, integrity, and secure access to cloud resources.

Platform Services and Real-Time Integration

The cloud platform provides several managed services that simplify application development and maintenance. Database services store customer, restaurant, and order information, while cloud storage services manage images and invoices. Notification services send real-time order confirmations, payment receipts, and delivery updates to customers through emails, SMS, or push notifications. Monitoring services continuously track application performance, server health, and resource utilization, allowing administrators

to detect and resolve issues quickly. Logging services maintain detailed records of user activities and system events, supporting troubleshooting and security auditing. The integration of these platform services enables smooth communication between frontend clients, backend applications, cloud storage, and infrastructure, ensuring real-time processing of customer requests.

System Workflow

The operation of the online food ordering platform begins when a customer accesses the application through a web browser or mobile app and logs into the system. The customer browses restaurant menus, selects food items, and places an order. The frontend application sends the order details to backend cloud services through RESTful APIs. The backend validates the authentication token, verifies menu availability, calculates the total payment, and records the order in the cloud database. Menu images and invoices are stored securely in cloud storage, while confirmation notifications are sent to both the customer and the restaurant. Restaurant staff prepare the order, and delivery personnel receive delivery information through their mobile application. Throughout the process, Kubernetes manages application scalability, HTTPS secures communication, and RBAC ensures that each user accesses only authorized system functions.

